

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN STACK

Practical – VII

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7.MongoDB Basics

a. Install and start MongoDB.

Steps:

1. Download **MongoDB Community Server** for Windows.
2. Run the .msi installer.
3. Select **Complete Installation**.
4. Click **Install** → **Finish**.
5. Open **MongoDB Compass**.
6. Connect to:

mongodb://localhost:27017

Output: MongoDB local server connected successfully.



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Community Server

The Community version of our distributed database offers a flexible document data model along with support for:

- Ad-hoc queries
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- Real-time aggregations for powerful ways to access and analyze your data

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Or Try MongoDB Atlas

Try our fully managed service, MongoDB Atlas, for free with capabilities designed to support you from prototype to production, including:

- Auto-scaling
- High availability by default
- Automatic backups
- Integrated full-text search, vector search, and stream processing
- Multi-region and multi-cloud support

Get Started For Free

Version

8.3.11 (current)

▾

Platform

Windows x64

▾

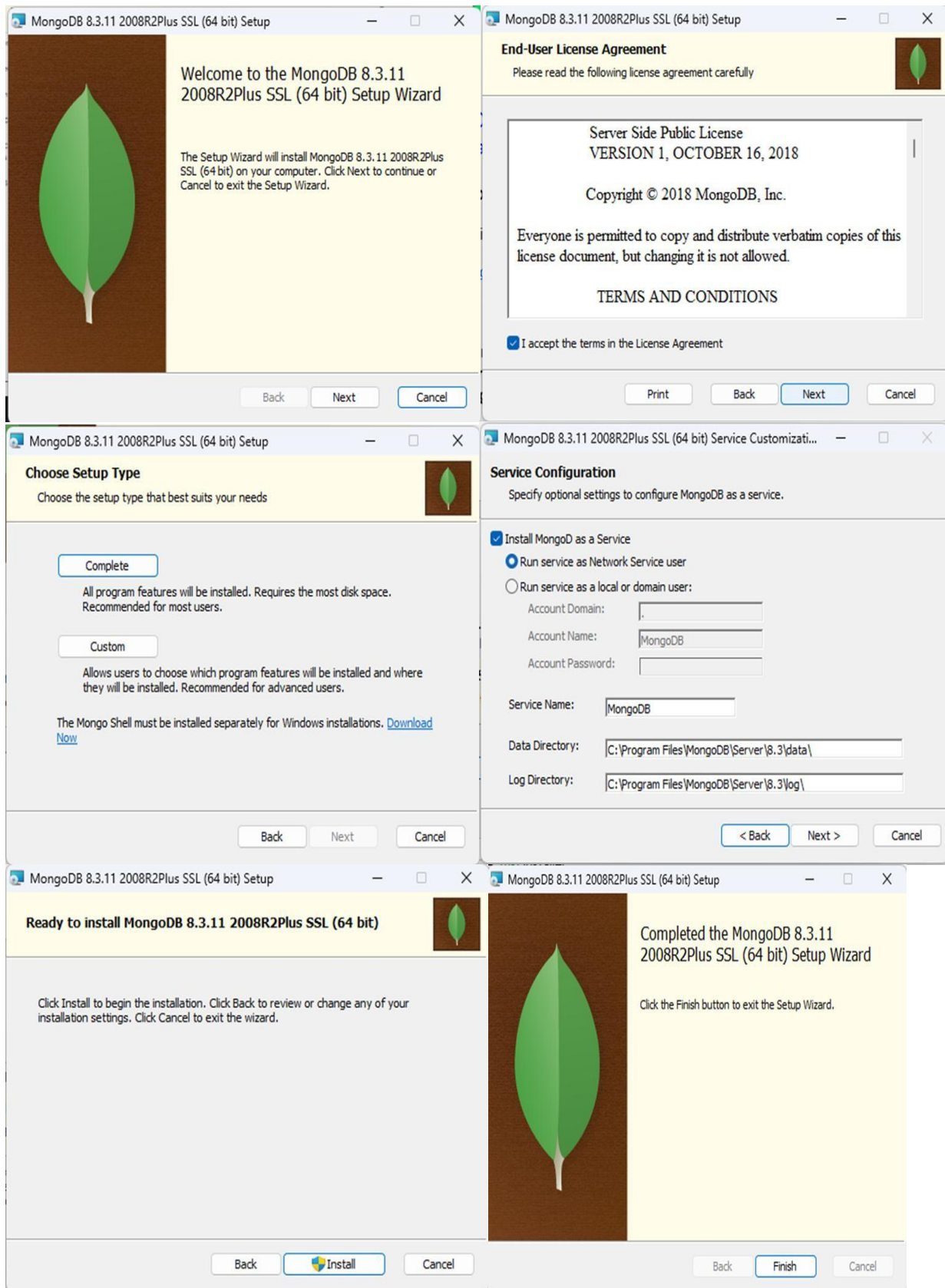
Package

msi

▾

Download 

More Options ...



b. Create a database and collection.

Steps:

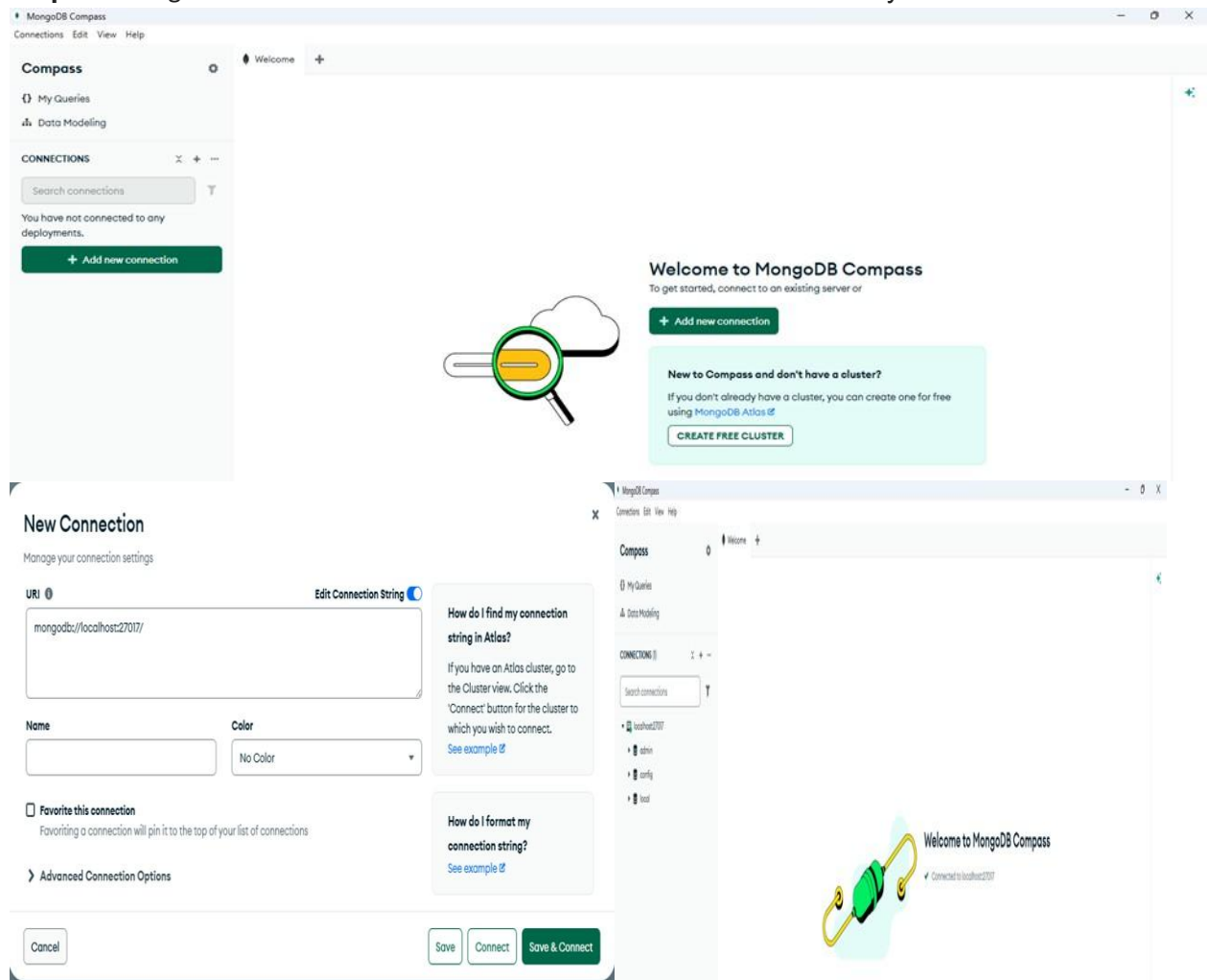
1. Open MongoDB Compass.
2. Click **+** beside the localhost connection.
3. Select **Create Database**.
4. Enter:

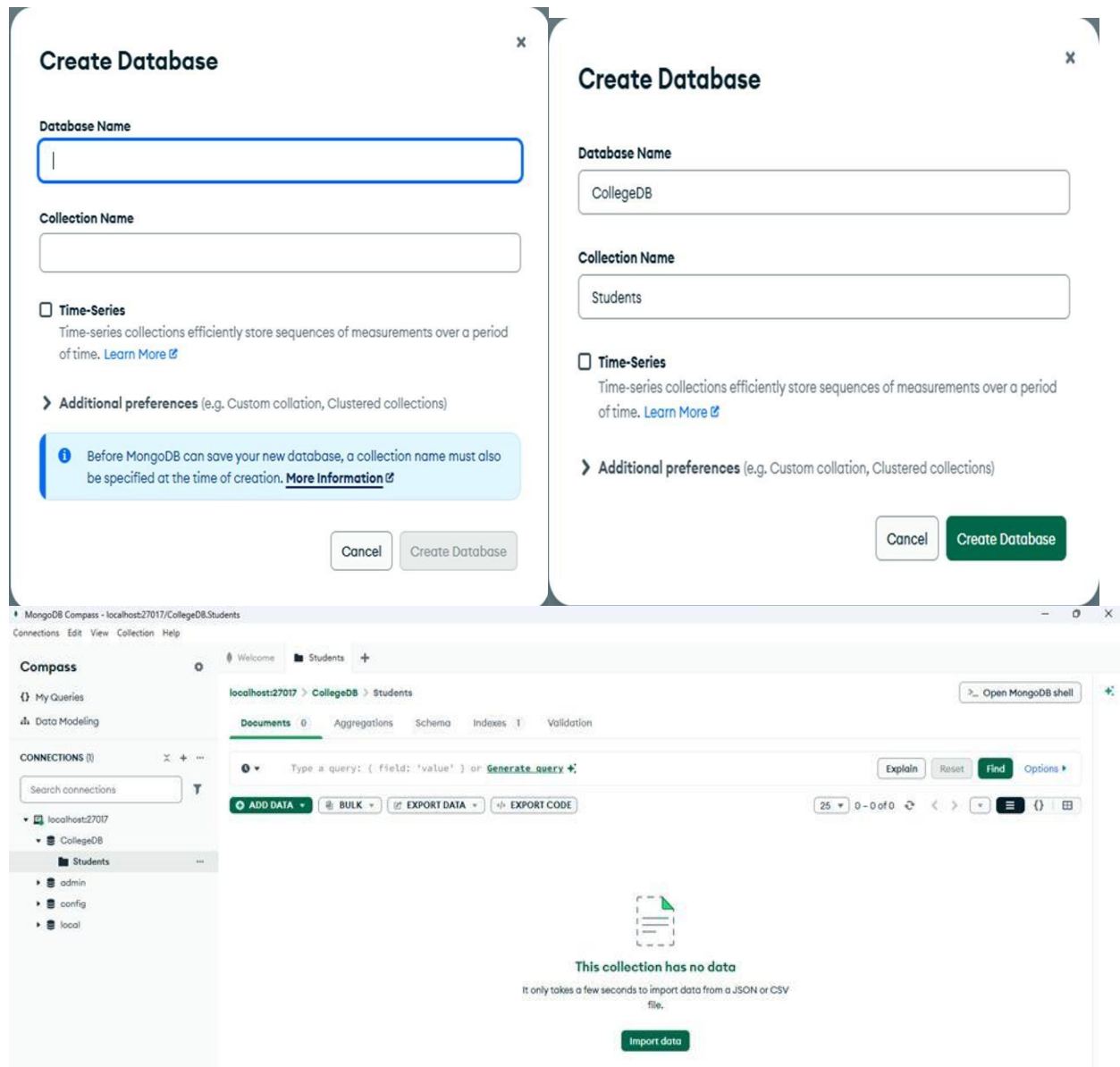
Database Name: CollegeDB

Collection Name: Students

5. Click **Create Database**.
6. Open CollegeDB → Students.
7. Click **ADD DATA** → **Insert Document**.

Output: CollegeDB database and Students collection created successfully.





c. Insert records in MongoDB.

Steps:

1. Open **MongoDB Shell (mongosh)**.
2. Select the database:

use CollegeDB

3. Insert one record:

```
db.students.insertOne({
  rollNo: 101,
  name: "Rahul",
  course: "B.Sc. IT",
  marks: 85
})
```

4. Insert multiple records: `db.students.insertMany([`

```
{rollNo: 102, name: "Priya", course: "B.Sc. IT", marks: 92},
{rollNo: 103, name: "Amit", course: "B.Sc. CS", marks: 78},
{rollNo: 104, name: "Neha", course: "B.Sc. IT", marks: 88}
```

])

Output: Records inserted successfully.

```
>_MONGOSH                                     > db.students.insertOne({
> use("CollegeDB")                             rollNo: 102,
< switched to db CollegeDB                     name: "Priya Patil",
> db["Students"].find()                       course: "B.Sc. IT",
< {                                             marks: 92
  _id: ObjectId('6aae298ba23fc5feddb02145'),
  rollNo: 101,
  name: 'Rahul Sharma',
  course: 'B.Sc. IT',
  marks: 85
}
}
CollegeDB>                                     < {
                                              acknowledged: true,
                                              insertedId: ObjectId('6aae2a27a6675fa2a5aad30a')
}

> db.students.insertMany([
  {
    rollNo: 102,
    name: "Priya Patil",
    course: "B.Sc. IT",
    marks: 92
  },
  {
    rollNo: 103,
    name: "Amit Joshi",
    course: "B.Sc. CS",
    marks: 78
  },
  {
    rollNo: 104,
    name: "Neha Kulkarni",
    course: "B.Sc. IT",
    marks: 88
  }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aae2a73a6675fa2a5aad30b'),
    '1': ObjectId('6aae2a73a6675fa2a5aad30c'),
    '2': ObjectId('6aae2a73a6675fa2a5aad30d')
  }
}
```

d. View records from the collection.**Steps:**

View all records: `db.students.find()`

View records in readable format:

`db.students.find().pretty()` **Find**

student with roll number 102:

`db.students.find({rollNo: 102})`

Find student with marks 92:

`db.students.find({marks: 92})`

Output: Required student records are displayed.

[Paste Screenshot Here]

e. Update and delete records.**Delete Record**

Delete student with roll number 104:

`db.students.deleteOne({rollNo: 104})`

Update Record

Change Rahul's marks from 85 to 90:

`db.students.updateOne(`

`{name: "Rahul"},`

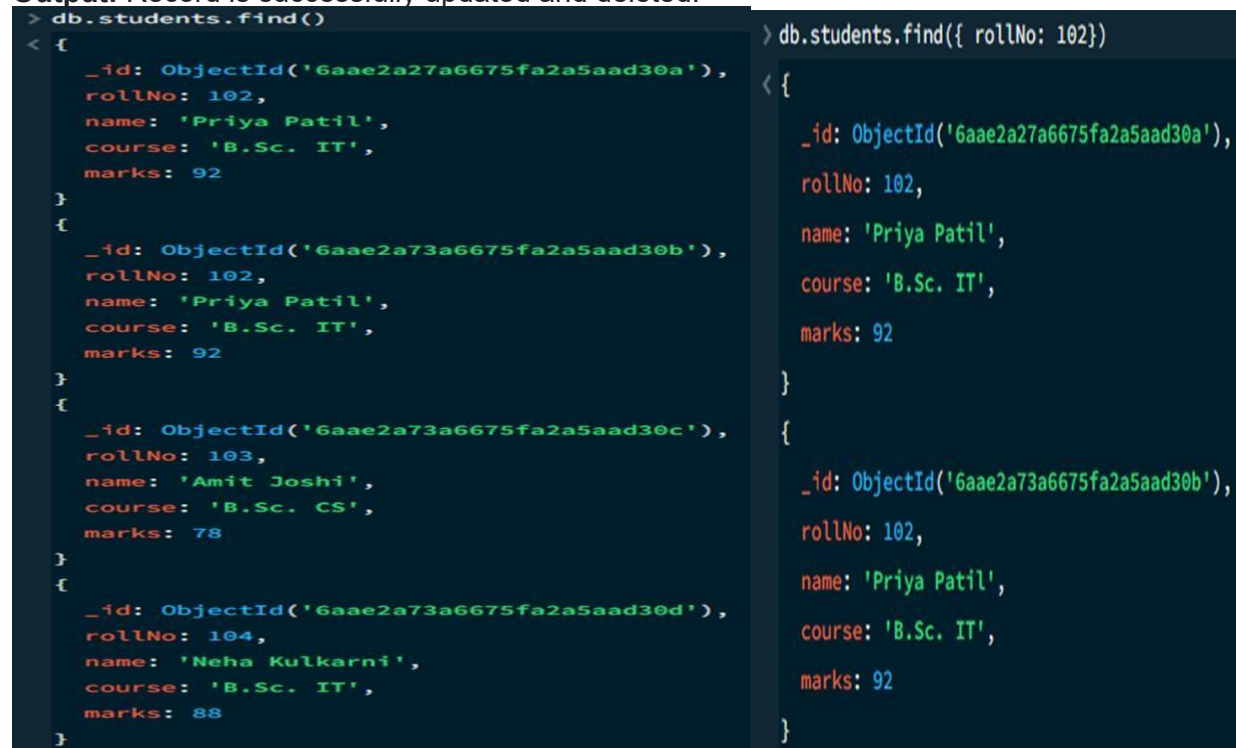
`{ $set: {marks: 90} }`

`)`

Verify Updated Record

`db.students.find({name: "Rahul"})`

Output: Record is successfully updated and deleted.



```

> db.students.find()
< {
  _id: ObjectId('6aae2a27a6675fa2a5aad30a'),
  rollNo: 102,
  name: 'Priya Patil',
  course: 'B.Sc. IT',
  marks: 92
}
{
  _id: ObjectId('6aae2a73a6675fa2a5aad30b'),
  rollNo: 102,
  name: 'Priya Patil',
  course: 'B.Sc. IT',
  marks: 92
}
{
  _id: ObjectId('6aae2a73a6675fa2a5aad30c'),
  rollNo: 103,
  name: 'Amit Joshi',
  course: 'B.Sc. CS',
  marks: 78
}
{
  _id: ObjectId('6aae2a73a6675fa2a5aad30d'),
  rollNo: 104,
  name: 'Neha Kulkarni',
  course: 'B.Sc. IT',
  marks: 88
}

> db.students.find({ rollNo: 102})
< {
  _id: ObjectId('6aae2a27a6675fa2a5aad30a'),
  rollNo: 102,
  name: 'Priya Patil',
  course: 'B.Sc. IT',
  marks: 92
}
{
  _id: ObjectId('6aae2a73a6675fa2a5aad30b'),
  rollNo: 102,
  name: 'Priya Patil',
  course: 'B.Sc. IT',
  marks: 92
}

```

```
> db.students.find({marks:92})
```

```
< {
  _id: ObjectId('6aae2a27a6675fa2a5aad30a'),
  rollNo: 102,
  name: 'Priya Patil',
  course: 'B.Sc. IT',
  marks: 92
}
{
  _id: ObjectId('6aae2a73a6675fa2a5aad30b'),
  rollNo: 102,
  name: 'Priya Patil',
  course: 'B.Sc. IT',
  marks: 92
}
```

```
> db.students.deleteOne({
  rollNo: 104
})
```

```
< {
  acknowledged: true,
  deletedCount: 1
}
```

```
> db.students.updateOne(
  { rollNo: 102 },
  { $set: { marks: 90 } }
)
```

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
> db.students.updateMany(
  { course: "B.Sc. IT" },
  { $set: { course: "Information Technology" } }
)
```

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

```
> db.students.updateMany(
  { marks: { $gt: 80 } },
  { $set: { grade: "A" } }
)
```

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

```
> db.students.find()
```

```
< {
  _id: ObjectId('6aae2a27a6675fa2a5aad30a'),
  rollNo: 102,
  name: 'Priya Patil',
  course: 'Information Technology',
  marks: 90,
  grade: 'A'
}
{
  _id: ObjectId('6aae2a73a6675fa2a5aad30b'),
  rollNo: 102,
  name: 'Priya Patil',
  course: 'Information Technology',
  marks: 92,
  grade: 'A'
}
{
  _id: ObjectId('6aae2a73a6675fa2a5aad30c'),
  rollNo: 103,
  name: 'Amit Joshi',
  course: 'B.Sc. CS',
  marks: 78
}
```

f. Connect MongoDB with Node.js.**Steps:**

1. Create and open a folder in **VS Code**.
2. Open Terminal.
3. Run:

```
npm init -y
```

4. Install Mongoose:

```
npm install mongoose
```

5. Create server.js.

6. Add the following code:

```
const mongoose = require("mongoose");
```

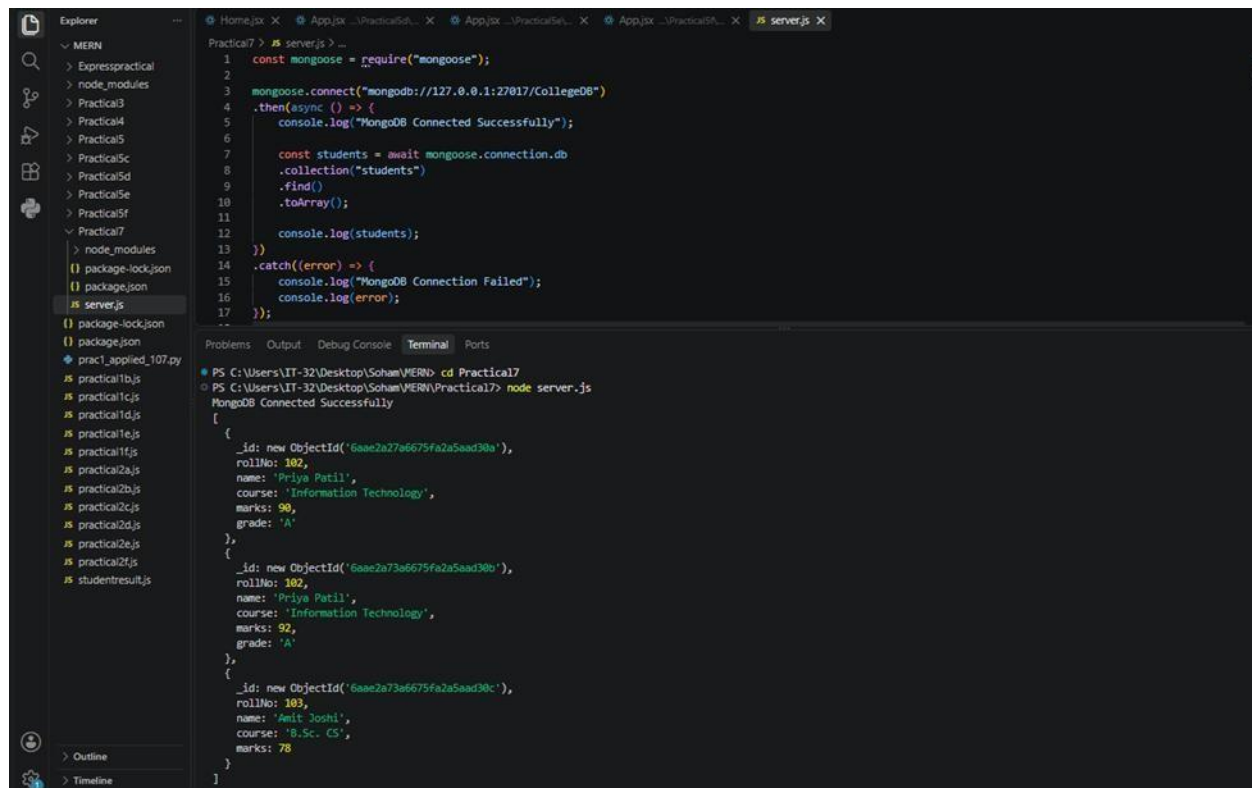
```
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
```

```
  .then(() => {  
    console.log("MongoDB Connected Successfully");  
  })  
  .catch((error) => {  
    console.log("Error:", error);  
  });
```

7. Run:

```
node server.js
```

Output: MongoDB Connected Successfully



```
1 const mongoose = require("mongoose");
2
3 mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
4   .then(async () => {
5     console.log("MongoDB Connected Successfully");
6
7     const students = await mongoose.connection.db
8       .collection("students")
9       .find()
10      .toArray();
11     console.log(students);
12   })
13   .catch((error) => {
14     console.log("MongoDB Connection Failed");
15     console.log(error);
16   });
17
```

```
PS C:\Users\IT-32\Desktop\Soham\VERN> cd Practical7
PS C:\Users\IT-32\Desktop\Soham\VERN\Practical7> node server.js
MongoDB Connected Successfully
[
  {
    _id: new ObjectId('6aae2a27a6675fa2a5aad30a'),
    rollNo: 102,
    name: 'Priya Patil',
    course: 'Information Technology',
    marks: 90,
    grade: 'A'
  },
  {
    _id: new ObjectId('6aae2a73a6675fa2a5aad30b'),
    rollNo: 102,
    name: 'Priya Patil',
    course: 'Information Technology',
    marks: 92,
    grade: 'A'
  },
  {
    _id: new ObjectId('6aae2a73a6675fa2a5aad30c'),
    rollNo: 103,
    name: 'Amit Joshi',
    course: 'B.Sc. CS',
    marks: 78
  }
]
```

Result

Successfully created a MongoDB database and collection, inserted, viewed, updated and deleted records, and connected MongoDB with Node.js using Mongoose.